

AI for Theory

State of the art, uses and actions for the community

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github.com/gpeyre/ia4maths

Talk roadmap

LLMs and mathematics

Problem 1

A line in the plane is called *sunny* if it is **not** parallel to any of the x -axis, the y -axis, and the line $x + y = 0$.

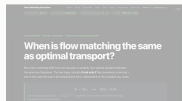
Let $n \geq 3$ be a given integer. Determine all nonnegative integers k such that there exist n distinct lines in the plane satisfying both of the following:

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AI for research in mathematics



Experience reports

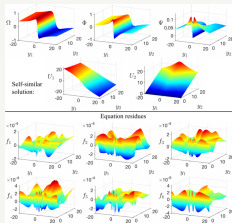


Impact on the community



AI for mathematics

AI for scientific discovery



PINNs to explore PDEs: Navier–Stokes / Euler, instabilities, counterexamples.

Focus: mathematical reasoning, from informal to formal.

Source: Wang–Lai–Gómez–Serrano–Buckmaster, arXiv:2201.06780 ; Hariharan et al., arXiv:2604.23468.

AI for theorem proving

$$A_8 := \left\{ (x_1, \dots, x_8) \in \mathbb{Z}^8 \mid \sum_{i=1}^8 x_i \equiv 0 \pmod{2} \right\}$$

The E_8 sphere packing $\mathcal{P}(A_8)$ is the packing of balls of radius $\frac{\sqrt{2}}{2}$ centred at points on A_8 . We define it in Lean as the following `PeriodicSpherePacking`:

```
noncomputable def ESPacking : PeriodicSpherePacking 8 where
  separation := √2
  lattice := E8Lattice
  centers := E8Lattice
  centers_dist := by ...
```

Viazovska's main result in [14] is the following:

Theorem 1 (Viazovska 2016). *There is no sphere packing in $\mathbb{R}^8$ with density greater than that of the E_8 packing. That is,*

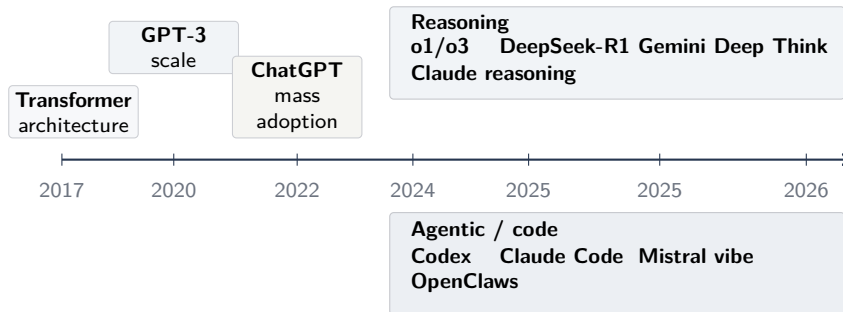
$$\Delta_k = \Delta(\mathcal{P}(A_8)) = \pi^4/384$$

In Lean, our statement of the main theorem is the above equality:

```
theorem SpherePacking_MainTheorem : SpherePackingConstant 8 =
  ESPacking_density := ...
```

Lean: verified proof; Viazovska's optimal E_8 packing.

LLMs for reasoning and mathematics



Why mathematics became central for LLMs

Problem : mathematics as a testbed for reasoning.

- ▶ **Training**: *next-token prediction*, evaluation oriented toward reasoning.
- ▶ **Benchmarks**: GSM8K, MATH, AIME/IMO-style as strategic signals.
- ▶ **Reasoning**: *reinforcement learning* and mathematical rewards.

Benchmark snippets

GSM8K

word arithmetic: “Alice buys 3 notebooks... how many are left?”

MATH

contest problem: algebra / geometry / analysis, structured solution.

AIME / IMO

short statement, long search: determine, optimize, prove.

Takeaway : agentic AI: local answer to full pipeline – plan, code, verify, write.

Source: Cobbe et al. 2021; Hendrycks et al. 2021; OpenAI Codex 2025/2026.

Simple problems: mathematical olympiads

Results 2024: formal reasoning

AlphaProof + AlphaGeometry 2, translation to formal languages then proof: **28/42**.

Results 2025: informal reasoning

Gemini Deep Think, general-purpose models and informal proofs: **35/42**.

Example of IMO 2025 statement

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Source: DeepMind 25/07/2024 and 21/07/2025; official IMO 2025 PDF (Deep Think).

Talk roadmap

LLMs and mathematics

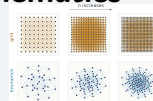
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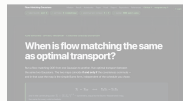
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AI for research in mathematics



Experience reports



Impact on the community



Research level: First Proof, from pilot to benchmark

Problem : reproducible measurement of research-level proving.

Task: unseen problems, available human solutions, full write-up; **criterion**: anonymized acceptance.

Results : First Batch → Second Batch

First Batch (02/2026)

Signal: ~2 clear successes (Q9,Q10).

Gray zone: Q5/Q8 close or repairable; no formal review.

Second Batch (03–06/2026)

OK: 17/39 submissions passed.

Coverage: 7/10 problems had ≥ 1 OK solution.

Criterion: flawless or minor revisions after anonymized expert review.

Takeaway : scale shift, but human validation remains central.

- ▶ **Capability**: serious proofs possible on nontrivial problems.
- ▶ **Costs**: batch 2, \$117–\$4.8k for 10 answers; ~\$10–\$480/answer.
- ▶ **Bottleneck**: expert review, skipped lemmas, “standard arguments”, fragile citations.
- ▶ **Benchmark**: prompts, logs, tokens, time, costs and acceptance criteria.

Source: [ressources/first-proof-summary.md](#) ; 1stProof ; [arXiv:2602.05192](#) ; [arXiv:2606.18119](#).

Research level: unit distance conjecture

Problem : $u(n)$, maximum number of unit-distance pairs among n points in the plane.

OpenAI prompt (summary)

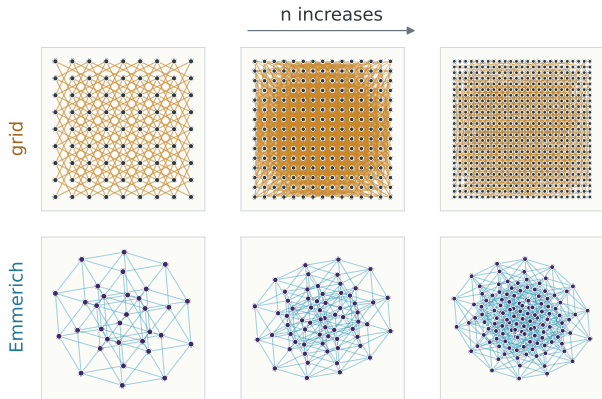
"Resolve Erdős's planar unit-distance problem completely."

Prove either $u(n) \leq n^{1+O(1/\log \log n)}$, or counterexamples to every bound $n^{1+C/\log \log n}$. **No partial progress.**

Results : OpenAI counterexample, then human verification and optimization.

	$1 + o(1)$	$\rightsquigarrow$	1.0152	
	Class.	OpenAI	Sawin	Emmerich
Exp.	$1 + o(1)$	> 1	1.014	1.0152

Takeaway : LLM + harness $\Rightarrow$ PhD-student level on an open problem.



Sources: OpenAI proof PDF p.3; Alon et al. 2605.20695; Sawin 2605.20579; Emmerich 2606.03419.

Formal mathematics: Viazovska theorem (Lean)

Problem : formalize an existing major theorem.

- ▶ **Object**: optimal packing in dimension 8 (Viazovska), extension to dimension 24.
- ▶ **Stakes**: transfer a very high-level proof into Lean.

Results : public Math, Inc. repository Sphere-Packing-Lean.

- ▶ **Status**: public, massive Lean project around certification of the E_8 / Leech packing proof.
- ▶ **Scope**: geometry of numbers, coding theory, 2022 Fields Medal.
- ▶ **Current size**: **830 files, 180 661 Lean lines**; FLT Lean reference: **117 files, 15 411 lines**.

Takeaway : large-scale formal certification, but sensitive governance.

- ▶ **Governance point**: academic (EPFL) → private (Math, Inc.) transfer was highly controversial.
- ▶ **Tension**: scientific credit, code maintenance, open Lean infrastructure.

Source: GitHub repositories; Hariharan et al., arXiv:2604.23468; local count 09/03/2026 with `rg + wc -l`.

Formal mathematics: analysis / PDEs (Lean)

Problem : formalize modern analysis, still only sparsely covered by `mathlib`.
Standard use case: math expert who is not a Lean specialist, guided by LLM + formal checking.

Results : **Scott Armstrong** (PDEs) and Julia Kempe;
Harnack, Hölder and reusable analysis foundations in
`scottnarmstrong/DeGiorgi`.
Short excerpt : `Harnack.lean`

```
theorem harnack
  (A : NormalizedEllipticCoeff d (ball 0 1))
  (hsol : IsSolution A.1 u) :
  essSup u  $\mu_{1/2} \leq \exp(C\_harnack d * A.1.\Lambda^{1/2}) * \text{essInf } u \mu_{1/2} := \text{by ...}$ 
```



Source: Armstrong–Kempe, arXiv:2604.05984; `scottnarmstrong/DeGiorgi`, especially `Harnack.lean`; Armstrong blog post 07/04/2026.

Takeaway

- ▶ **Feasibility**: modern analysis with Claude/Codex pro accounts.
- ▶ **Expertise**: math guidance essential; impossible outside the domain.
- ▶ **Cost**: many tokens, manageable with a solid blueprint.
- ▶ **Tension**: math expert vs Lean idiom / core developers.
- ▶ **Goal**: formalized foundations to put Lean in the loop.

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AI for research in mathematics



Experience reports



Impact on the community



My research experience

Problem : make progress on a fairly well-scoped problem where one is stuck; use Lean as a non-specialist.

Results : personal report from intensive use.

- ▶ **Unlock**: open problem solved in **2 weeks** through intensive GPT-5 prompting.
- ▶ **Preprint**: <https://arxiv.org/abs/2602.01372>
- ▶ **Agents**: Codex / Claude Code in daily use.
- ▶ **Library**: <https://github.com/gpeyre/flow-sinkhorn>

Takeaway : two regimes and workflow gains.

- ▶ **Informal**: unlocks open problems.
- ▶ **Formal (Lean)**: works, but costs $\sim 10\times$ more tokens.
- ▶ **Writing**: faster prototyping.
- ▶ **Technical**: automated repetitive tasks.
- ▶ **Reasoning**: Lean certifies the proof without directly improving the mathematical idea.

Source: Personal experience + arXiv/GitHub (accessed 09/03/2026).

arXiv:2602.01372v1 [math.OC] 1 Feb 2026

Robust Sublinear Convergence Rates for Iterative Bregman Projections

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February 3, 2026

Abstract

Entropic regularization provides a simple way to approximate linear programs whose constraints split into two (or more) tractable blocks. The resulting objectives are amenable to cyclic Kuhn-Munkres (KL) Bregman projections, with the classical Sinkhorn algorithm for optimal transport (balanced, unbalanced, gradient flows, barycenters, ...) as the canonical example. Assuming uniformly bounded primal sums and dual values, we prove that the dual objective of these KL projections decreases at an $O(1/k)$ rate with a constant that scales only linearly in $1/\gamma$, where γ is the entropic regularization parameter. This extends the guarantees known for entropic-optimal transport to any such linearly constrained problem. Following the terminology introduced in [15], we call such rates “robust”, because this mild dependence on γ underpins broadly-complexity bounds for approximating the unregularized problem via alternating KL projections. The crucial aspect of the analysis is that the dual values should be measured according to a block-quotient dual norm, which depends on the structure of the split of the constraint into blocks. As an application, we derive the flow-Sinkhorn algorithm for the Wasserstein-1 distance on graphs. It achieves ϵ -additive accuracy on the transshipment cost in $O(p/\epsilon^2)$ arithmetic operations, where p is the number of edges.

1 Introduction

The aim of this paper is to provide a blueprint for the analysis of iterative Bregman projection onto affine spaces. The resulting convergence rates are sublinear (and therefore slow), but the associated constants scale favorably with the dimension, which is essential for using these methods to discretize and approximate high-dimensional problems. The analysis is designed with applications to Optimal Transport in mind, in particular on graph structures, where such guarantees were not previously available.

1.1 Entropic regularization of structured linear programs

We consider a feasible linear program

$$\max_{A \in \mathbb{R}^{k \times m}} \langle x, c \rangle \quad (1)$$

where constraints $Ax = b$ naturally split into blocks. We focus on the two-block case for clarity. Throughout, let $d \in \mathbb{N}$ be the dimension of the problem and write B_d^1 for the positive unit ball over

1

Group experience report

Problem : collective evaluation at CSD (ENS).

- ▶ **Protocol**: same prompt, multiple answers.
- ▶ **Task**: semi-open question, notebook, code, paper, formalization.
- ▶ **Audit**: submissions evaluated across several criteria.

Prompt excerpt

Consider flow matching with a stochastic interpolant $a(t)*X_0+b(t)*X_1$.
In python/ do an indepth numerical simulation ... $X_0 \sim N(0, Id)$,
 X_1 is a mixture of three Dirac. In paper/ write a detailed LaTeX
article ...
compute in closed form Σ_t and the flow map T_t .

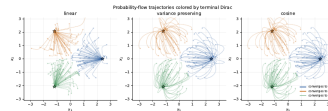


Figure 2: Sample trajectories of the flow-matching ODE for three schedules. Colors indicate the Dirac mass to which the final point is closest.

4 Gaussian Endpoint Geometry

We now take independent Gaussian endpoints

$$X_0 \sim \mathcal{N}(0, \Sigma_0), \quad X_1 \sim \mathcal{N}(0, \Sigma_1),$$

where Σ_0, Σ_1 are symmetric positive definite matrices.

Theorem 1 (Gaussian stochastic-interpolant flow). Let $X_t = a(t)X_0 + b(t)X_1$ with independent centered Gaussian endpoints. Define

$$\Sigma_t = a(t)^2 \Sigma_0 + b(t)^2 \Sigma_1, \quad A = \Sigma_0^{-1/2} \Sigma_t \Sigma_0^{-1/2},$$

and

$$M_t = a(t)^2 I + b(t)^2 A.$$

Then $X_t \sim \mathcal{N}(0, \Sigma_t)$, and the conditional velocity is linear,

$$u_t(x) = B_t x, \quad B_t = (a(t)\dot{a}(t)\Sigma_0 + b(t)\dot{b}(t)\Sigma_1)\Sigma_t^{-1}. \quad (5)$$

The flow map from time 0 to time t is

$$T_t = \Sigma_0^{1/2} M_t^{1/2} \Sigma_0^{-1/2}. \quad (6)$$

Snapshot: paper page from the codex-gabriel/ submission.

Source: ressources/agentic-benchs/todo.md; comparative report from 18/06/2026; codex-gabriel submission.

Group experience report (continued)

Problem : multicriteria evaluation.

Results : LLM-as-judge aggregation

Submission	Math	Code	Overall
emmanuel	9.3	8.8	9.1
codex-gabriel	9.0	9.0	8.8
openclaw	8.2	8.6	8.2
hermes	7.8	8.0	7.7
claude-gabriel	6.5	7.5	7.1
codex-kimia	7.0	7.0	7.0
codex-clement	7.0	5.5	6.2
antigravity-a.	5.8	6.0	5.8
antigravity-o.	5.5	5.0	5.4

Overall = aggregation of math, code, paper and robustness criteria.

Source: CSD experiment; audit [ressources/agentic-benchs/report](#).

Results : robust signals

- ▶ **Math**: Codex $\gg$ Claude $\gg$ rest.
- ▶ **Code**: Claude $\gg$ Codex $\gg$ rest.
- ▶ **Budget**: performance $\simeq$ tokens.

Takeaway : effect of token budget

Budget	Typical outcome
Small	incorrect proof
Medium	proof OK, statement imperfect
Large	complete solution
Huge	Lean viable

Zoom: Emmanuel Sérié / Noogram

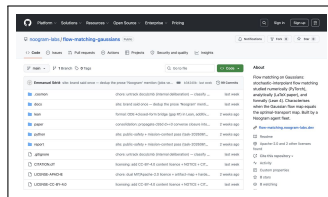
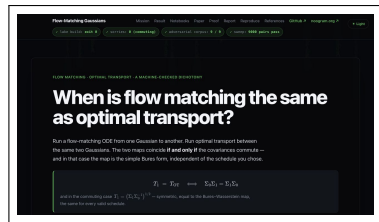
Problem : CSD prompt → full scientific artifact.

Results : Noogram contribution

- ▶ **Theorem**: Gaussian FM = OT iff covariances commute.
- ▶ **Artifacts**: site, code, paper, figures, Lean, report.
- ▶ **Audit**: false proof found and fixed; score **9.1**.

Takeaway : agent harnessing for science.

- ▶ **Chain**: generate, test, refute, publish.
- ▶ **Rigor**: Lean as anchor; expert required.
- ▶ **Cost**: heavy, but auditable end to end.



Sources: Noogram site; GitHub noogram-labs/flow-matching-gaussians; CSD audit.

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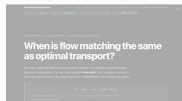
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AI for research in mathematics



Experience reports



Impact on the community



Impact on evaluation and teaching

Reviewing: AI as first reader



- ▶ **Automation:** first report, local flaws, suggestions.
- ▶ **Weak signal:** triage, not scientific validation.
- ▶ **Human value:** validity, novelty, taste.

Source: PaperReview.ai, Stanford ML Group (screenshot 02/07/2026) ; qualitative synthesis.

Teaching: preserve human training

- ▶ **Level:** PhD-level writing, code, local critique.
- ▶ **Risk:** delegating before learning to reason.
- ▶ **Loss:** future teachers, reviewers, AI evaluators.
- ▶ **Protect:** oral exams, proof critique, AI audit.

Actions for the community

Structure the community



- ▶ **CNRS RT**: bridge AI, math, formalization, CS (INSMI/INS2I).
- ▶ **AISSAI**: “AI for mathematics” quarter with the RT.

Open access to models

- ▶ **Repos**: academic projects connected to models.
- ▶ **Models**: control and auditability.

Dialogue with industry

- ▶ **DeepMind**: funds AISSAI postdoc program.
- ▶ **Mistral**: Emmy tools for CNRS to develop further.

Do not forget training

- ▶ **Faculty**: design + evaluation.
- ▶ **Evaluators**: preserve human training.

Conclusion

1. Acceleration already visible

- ▶ **Performance:** very fast progress.
- ▶ **Workflows:** code, verify, write.

2. Concrete scientific impact

- ▶ **Informal:** open problems unlocked.
- ▶ **Formal:** Lean progressing, high token cost.

3. Human role displaced

- ▶ **Positioning:** augment, not replace.
- ▶ **Role:** researcher → judge of proof quality.

4. Collective stakes

- ▶ **Access:** controllable and auditable models.
- ▶ **Tension:** academic–industry exchanges.
- ▶ **Sovereignty:** scientific and institutional framing.